DreamEEG: A Hybrid CNN–Transformer Decoder for Visual-Imagery Brain–Computer Interfaces in ALS and Locked-In Syndrome

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***Abstract*—Amyotrophic lateral sclerosis (ALS) and locked-in syndrome can leave an intact mind trapped inside a still body, so that a brain-based channel becomes the only remaining way to communicate. Most electroencephalography (EEG) based brain–computer interfaces (BCIs) rely on motor imagery, yet that pathway routes through exactly the motor system ALS destroys. Visual imagery (VI), mentally picturing an object, is a motor-free and intuitive alternative, but its neural signatures are weak and noisy. A recently released VI dataset (Gao *et al.*, 2026) exposes the core problem: the strongest published baseline (EEGNet) reaches only about 75% accuracy on figures and animals and just 62% on the object category, and only in a within-subject setting. This midterm report frames the decoder itself as the bottleneck and proposes DreamEEG, a lightweight hybrid convolutional–Transformer network. A shared multi-scale convolutional front-end feeds a variance-coupled dual-stream encoder, in which a first-order (mean) stream and a second-order (log-variance / band-power) stream are each refined by gated dual-attention Transformer blocks and then fused for a three-class within-category decision. We describe the dataset, the motivation, the proposed architecture and hypothesis, and a staged evaluation plan whose primary target is to beat the EEGNet baseline while closing the weak object-category gap and improving cross-session and cross-subject stability.**

***Index Terms*—Brain–computer interface, EEG, visual imagery, deep learning, convolutional neural network, Transformer, attention, ALS, locked-in syndrome, assistive technology.**

I. Introduction

AMYOTROPHIC lateral sclerosis (ALS) and the locked-in syndrome that can follow it progressively strip away voluntary movement while typically leaving cognition intact. The result is a patient who is fully aware but unable to speak, gesture, or type. For such individuals a brain-based communication channel is frequently the *only* avenue left, which has made non-invasive brain–computer interfaces (BCIs) a long-standing goal of assistive neurotechnology [1], [2].

The dominant control paradigm for electroencephalography (EEG) based BCIs is *motor imagery*, in which the user imagines moving a limb. This is a fundamental limitation for the ALS population: motor imagery is decoded from signals generated by the sensorimotor system, and that system is precisely what ALS degrades. A control strategy that does not depend on the motor pathway is therefore highly desirable.

Visual imagery (VI), deliberately picturing an object such as a dog, a fish, or a bird, offers exactly such a motor-free route. It is intuitive for users and does not require residual movement. The difficulty is that the cortical signals produced by silent visualization are weak, low in signal-to-noise ratio, and spatially diffuse, which makes them hard to decode reliably.

A newly published, openly available VI dataset [3] has, for the first time, made systematic VI decoding research possible on a shared benchmark. However, the accompanying baseline results reveal that decoding is not yet good enough for real use: a compact convolutional network (EEGNet [4]) achieves roughly 75% on the figure and animal categories but only about 62% on objects, and only when trained and tested on the *same* subject. For a system that may one day be a person's sole voice, this level of accuracy and generalization is insufficient.

The real gap: We identify the **decoder** as the bottleneck. The signal exists and has been recorded carefully; what is missing is a model architecture that extracts the discriminative structure of VI activity robustly enough, especially for the hardest object category and beyond a single calibrated user. This report presents the problem, the data, and the DreamEEG architecture we set out to build to close that gap.

Contributions: This midterm report makes three contributions:

- a clear formulation of VI decoding for ALS/locked-in communication and of the specific accuracy and generalization gap left open by the current baseline;
- the design of DreamEEG, a lightweight hybrid CNN–Transformer with a novel variance-coupled dual-stream encoder aimed at the weak object category; and
- a staged experimental and evaluation plan, from baseline reproduction through cross-session and cross-subject transfer.

II. Problem Statement and Motivation

The motivation for this work can be summarized in four points.

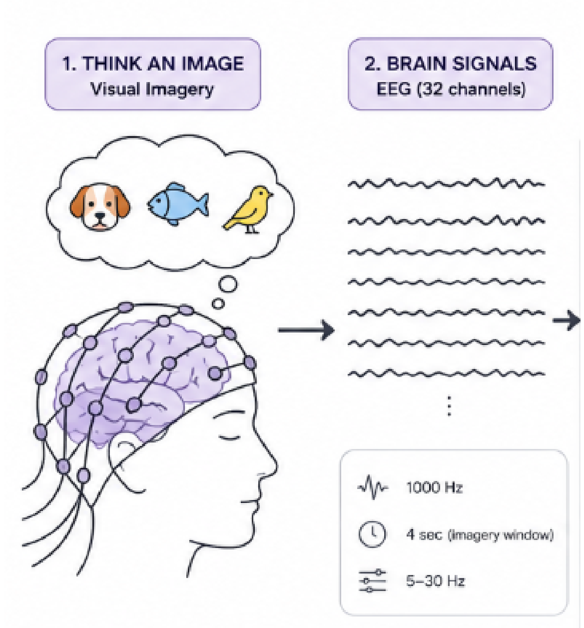

Fig. 1. Concept of visual-imagery decoding. The user mentally pictures an object (e.g., dog, fish, bird); 32-channel EEG is recorded at 1000 Hz and band-pass filtered to 5–30 Hz; the decoder classifies the 4 s imagery window.

1) **Communication of last resort.** ALS and locked-in syndrome trap an intact mind in a still body; a brain-based channel is often the only way left to communicate.
2) **Motor imagery is the wrong pathway.** Most EEG-BCIs rely on motor imagery, but that routes through the motor system, exactly what ALS destroys.
3) **Visual imagery is promising but hard.** VI is a motor-free, intuitive path, yet the neural signals are weak and noisy.
4) **The data exists, the decoding does not.** A new VI dataset exists [3], yet decoding remains weak: the best baseline (EEGNet) reaches only about 75%, and just 62% on objects.

The real gap. Decoding is not yet accurate or reliable enough to be someone's only voice. The object category sits at just 62%, and this has been tested only *within* the same person. Because the bottleneck is the decoder itself, that is what we set out to improve.

Fig. 1 illustrates the end-to-end idea: a user pictures one of a small set of objects, 32-channel EEG is recorded during a fixed imagery window, and a decoder must recover which object was imagined.

III. Related Work

We organize prior work into four strands: decoding backbones, visual-imagery and semantic decoding, cross-subject and transfer learning, and clinical BCIs for ALS and locked-in syndrome.

A. Decoding Backbones

EEGNet [4] is a compact CNN using depthwise and separable convolutions that generalizes across many EEG paradigms (P300, error potentials, motor imagery, SSVEP); it is the deep model used in our dataset's own validation and therefore our natural baseline. Classical pipelines are well surveyed by Lotte *et al.* [5], whose review of feature extraction and classification (CSP, LDA, SVM, k-NN) frames the CSP+KNN baseline that the dataset also reports. The EEG Conformer [6] combines a convolutional front-end with self-attention and reached state-of-the-art results on motor-imagery and emotion tasks, making it a strong candidate backbone. EEG-Deformer [7] adds a hierarchical coarse-to-fine temporal module and an information-purification block, matching or beating the state of the art on attention, fatigue, and workload tasks while producing neurophysiologically meaningful attention maps. Most relevant to our lightweight goal, DBConformer [8] models temporal and inter-channel dependencies in separate branches with a channel-attention module and outperforms ten or more baselines using roughly eight times fewer parameters than the original Conformer. DreamEEG builds on this CNN + Transformer lineage but adds an explicit variance-coupled dual-stream design.

B. Visual-Imagery and Semantic Decoding

The dataset of Gao *et al.* [3] is, to our knowledge, the first open, VI-focused benchmark of its scale, and its EEGNet baseline (75% / 62% on objects, within-subject) sets the bar for this work. The breadth of the imagery-decoding field is illustrated by EEG2Video [9], which reconstructs dynamic visual perception from EEG. For imagery specifically, Wilson *et al.* [10] released an open 128-channel semantic-concept dataset used as a comparison point by our target dataset, and Pan *et al.* [11] pushed VI toward spelling by classifying 29 characters from visual imagery. Kilmarx *et al.* [12] systematically evaluated VI as a control strategy and reported that it is feasible but limited by high variability and modest accuracy, precisely the motivation for stronger decoders. Lee *et al.* [13] established VI and imagined speech as intuitive, endogenous paradigms, and Sousa *et al.* [14] showed that pure VI alone can provide three classes of control for non-motor disorders. On the neuroscience side, Rybar *et al.* [15] showed imagined semantic categories are decodable from simultaneous EEG and fNIRS, and Huang *et al.* [16] used a spatio-temporal capsule network, arguing that semantic decoding generalizes better than low-level visual-feature decoding.

C. Cross-Subject and Transfer Learning

Because the practical bottleneck for VI-BCIs is per-user calibration, transfer learning is central to our staged plan. Wang *et al.* [17] built a lightweight semantic-decoding network that explicitly balances group-level commonality against individual variability via group-to-individual transfer, the closest published idea to our own generalization goal. Junqueira *et al.* [18] showed that Euclidean Alignment is a simple, low-cost normalization that reliably improves cross-subject decoding with deep networks, which is why we adopt it as a first alignment step. Wu *et al.* [19] provide a tutorial that

TABLE I
SUMMARY OF THE VISUAL-IMAGERY EEG DATASET [3]

Property	**Value**
Source	Gao *et al.* (2026), *Scientific Data*; open; VI-focused; new (no prior decoding work)
Participants	22 healthy; aged 20–23; 5 female; two sessions on different days (19 of 22 completed both)
Recording	32-channel wireless system; 1000 Hz; 10–20 layout (AFz ground, CPz reference); shielded room
Stimuli	10 images; 3 categories (figures, animals, objects)
Trial (17 s)	fixation 3 s $\rightarrow$ image 4 s $\rightarrow$ mask 2 s $\rightarrow$ imagery 4 s $\rightarrow$ rest; the 4 s imagery window is decoded
Task	3-class within a category (e.g., dog vs. fish vs. bird); chance = 33%; 120/120/160 trials
Preproc.	5–30 Hz band-pass; common average reference; ICA (Picard) + ICLabel; 4 s epochs
Quality	open BIDS format; raw + preprocessed; baseline code (MNE, PyTorch)

organizes EEG transfer learning into cross-subject and cross-session settings and into instance-, feature-, and parameter-based methods, giving us a menu of adaptation strategies.

D. Clinical BCIs for ALS and Locked-In Syndrome

Birbaumer *et al.* [1] gave the first proof that paralysed users, including people with ALS, could spell by self-regulating slow cortical potentials with only a scalp EEG cap. Kübler *et al.* [20] showed people severely disabled by ALS could control a sensorimotor-rhythm BCI, and Nijboer *et al.* [21] took a P300 speller into ALS patients' homes with accuracy sustained over months. Moving to invasive systems, Vansteensel *et al.* [2] let a locked-in ALS patient type at home through a fully implanted BCI, and Willett *et al.* [22] decoded attempted speech at about 62 words per minute. These systems are powerful but invasive and heavily per-user calibrated; our work targets a non-invasive, lightweight alternative built on visual rather than motor or speech imagery.

IV. DATASET

All experiments use the openly released visual-imagery EEG dataset of Gao *et al.* [3], published in *Scientific Data*. Its key properties are summarized in Table I.

The decoding target is deliberately narrow and well defined: a three-way, within-category classification of the 4 s imagery epoch, giving a 33% chance level. Recordings span two sessions on different days, which lets us measure day-to-day stability, and the open BIDS packaging with reference code makes faithful baseline reproduction possible. The dataset and its descriptor paper are publicly available.[1]

Fig. 2 shows the general non-invasive EEG acquisition setup that underlies this recording paradigm.

[1]Dataset: https://figshare.com/articles/dataset/EEG_Dataset_for_Visual_Imagery/30227503; descriptor: https://www.nature.com/articles/s41597-025-06512-5.

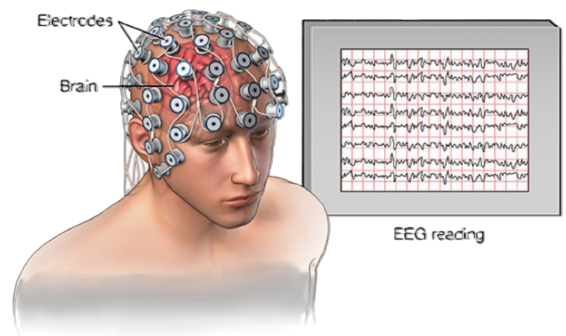

Fig. 2. Non-invasive scalp EEG acquisition: an electrode cap records cortical activity that is digitized into multi-channel time series (the "EEG reading"). The study dataset uses a 32-channel, 10–20 montage.

V. RESEARCH OBJECTIVES

The project pursues six objectives, ordered from the core technical goal to its clinical justification.

1) **Stronger decoder (primary).** Design a combined CNN–Transformer that beats the EEGNet baseline.
2) **Fix the weakest task.** Lift object-category accuracy ($\sim$62%) toward the level of figures and animals ($\sim$75%).
3) **Lightweight and practical.** Keep few parameters, train on a single GPU, and remain deployable for real assistive use.
4) **Stable over time.** Stay consistent from day to day (session 1 $\rightarrow$ session 2).
5) **Generalize to new users.** Reduce per-user calibration via transfer learning and alignment.
6) **Clinical relevance.** Be reliable enough to aid ALS / locked-in communication.

VI. PROPOSED METHOD

A. Hypothesis

Our working hypothesis is twofold. First, not all electrodes contribute equally to VI decoding; we therefore begin by identifying which probes/channels are most informative for this specific problem. Second, on top of that channel evidence, a *customized* hybrid model that jointly models first- and second-order signal statistics will extract more discriminative structure than a single-stream CNN, and will do so most where the baseline is weakest (the object category).

B. Architecture

DreamEEG (Fig. 3) takes a preprocessed EEG epoch of 32 channels $\times$ 4 s, band-pass filtered to 5–30 Hz, and passes it through three stages.

a) Shared front-end: A *multi-scale temporal convolution* with parallel kernels of length 32, 64, 96, and 128 captures rhythmic structure at several time scales simultaneously. A *depthwise spatial convolution* ($C \times 1$), followed by batch normalization and an ELU nonlinearity, then learns channel/spatial filters analogous to spatial patterns.

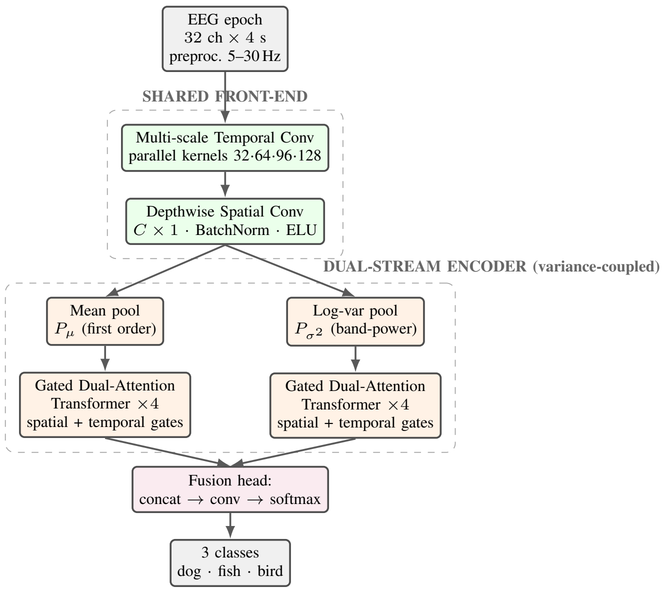

Fig. 3. The proposed DreamEEG architecture: a shared multi-scale convolutional front-end, a variance-coupled dual-stream encoder pairing a first-order (mean) and a second-order (log-variance / band-power) path, each refined by gated dual-attention Transformer blocks, and a fusion head producing the 3-class decision.

b) Variance-coupled dual-stream encoder (key edge): The shared features branch into two complementary streams:

- a **first-order stream** using a mean pool (P_μ) that summarizes average activation, and
- a **second-order stream** using a log-variance pool (P_{σ^2}) that captures band-power / signal energy, a feature long known to be informative for imagery EEG.

Each stream is refined by a stack of four *gated dual-attention Transformer* blocks that apply both spatial and temporal attention gates, letting the model weight informative channels and time points.

c) Fusion head: The two stream representations are concatenated, passed through a convolutional layer, and mapped by a softmax to the three within-category classes (e.g., dog, fish, bird).

The design keeps the parameter count modest, in the spirit of compact decoders such as EEGNet and DBConformer, so that it remains single-GPU trainable and deployable, in line with Objective 3.

VII. Evaluation Protocol and Staged Plan

A. The Bar to Beat

The primary benchmark is the EEGNet baseline reported with the dataset [3], evaluated within-subject on the 3-class within-category task (chance = 33%). Table II lists the per-category accuracies. Objects, at 62.0%, are the clear weak spot and are therefore our main improvement target: the hardest but most useful case.

Evaluation uses the same 22 subjects, 32 channels at 1000 Hz, and two sessions provided in the open BIDS release with baseline code, so that our numbers are directly comparable.

TABLE II
EEGNet Baseline to Beat (Within-Subject) [3]

Category	**Accuracy**	**Chance**
Figures	75.1%	33%
Animals	75.8%	33%
Objects	**62.0%**	33%

B. Staged Plan

The plan is deliberately staged so that each step yields a usable result even if a later step underperforms, which keeps risk low against the project timeline.

1) **Reproduce the baseline.** Re-run the authors' within-subject EEGNet and CSP+KNN to confirm our pipeline matches their reported numbers, validating that we read and preprocess the data correctly.
2) **Build and tune DreamEEG.** Develop the hybrid CNN–Transformer, our core contribution, and tune it to beat the within-subject baseline, with particular attention (and augmentation) on the weak object category.
3) **Cross-subject benchmark.** Move to leave-one-subject-out (LOSO) evaluation (train on 21 subjects, test on the held-out one) and add Euclidean Alignment [18] as a first, low-cost step to close the expected generalization gap. No cross-subject benchmark exists on this dataset yet, so quantifying it is itself a contribution.
4) **Transfer learning and cross-session test.** Apply light domain adaptation / fine-tuning [19] to measure how few target trials are needed for acceptable accuracy, and train on session 1 / test on session 2 (19 two-session subjects) to report temporal stability.

Throughout, we report accuracy, macro-F1, and Cohen's κ, together with per-category results and confusion-matrix structure, in the within-subject, cross-session, and cross-subject (LOSO) settings, alongside the parameter count to document the lightweight goal.

VIII. Expected Outcomes and Conclusion

We expect DreamEEG to deliver a stronger yet lighter visual-imagery decoder that surpasses the EEGNet baseline, with the largest gains on the currently weak object category, while remaining stable across sessions and more transferable across users. If achieved, such a decoder moves VI-based BCIs a concrete step closer to being a dependable communication channel. For a person with ALS or locked-in syndrome, a brain–computer interface could then actually help restore a voice.

This midterm report has established the clinical motivation, isolated the decoder as the true bottleneck, described the open VI dataset and its baseline, and specified the DreamEEG architecture together with a staged plan to validate it. The next phase focuses on baseline reproduction and the first full training runs of the proposed model.

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